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Education

2019 **PhD in Marine Geophysics**, MIT-WHOI Joint Program
2014 **BA Physics, BS Geophysical Sciences**, University of Chicago

Professional Appointments

2025 – on **Lamont Assistant Research Professor**, Lamont-Doherty Earth Observatory, Columbia University
2025 – on **Adjunct Scientist**, Woods Hole Oceanographic Institution
2025 – 2025 **Research Associate 3**, Woods Hole Oceanographic Institution
2021 – 2025 **Postdoctoral Investigator**, Woods Hole Oceanographic Institution
2019 – 2021 **Fossett Postdoctoral Fellow**, Washington University in St Louis

Publications

2026 Tominaga, M, Tivey, M, **Mark, HF**, Sager, W, Preine, J. New Phoenix Magnetic Anomaly Data Confirms Global Nature of Jurassic Quiet Zone and Provides Insight into Early Pacific Spreading. *Geophysical Research Letters*. doi:[10.1029/2025GL117569](https://doi.org/10.1029/2025GL117569)

2025 **Mark, HF**, Wiens, DA, Ben Mansour, W, Zhou, Z. Depth-varying azimuthal anisotropy and mantle flow in the Patagonian slab window. *Seismica*. doi:[10.26443/seismica.v4i2.1670](https://doi.org/10.26443/seismica.v4i2.1670)

2024 **Mark, HF**. Refining the extent and depth of the shear zone surrounding the Alpine Fault using receiver function harmonics. *Geophysical Research Letters*. doi:[10.1029/2024GL112092](https://doi.org/10.1029/2024GL112092)
Preprint [available on essoar](#)

2024 **Mark, HF**, Zhu, J, Tominaga, M, Aliod, D, Tivey, M. shipgrav: A Python package for marine gravimetry. *Journal of Open Source Software*. doi:[10.21105/joss.07358](https://doi.org/10.21105/joss.07358)
Open code: [github repository](#), [documentation](#), and [archive of v1.0.6](#)

2024 Gourdeau, A, Prush, VB, Rowe, C, Nackers, C, **Mark, HF**, Morris, I, Rosset, P, Lamothe, M, Chouinard, L, Tarling, MS. Investigation of suspected Holocene fault scarp near Montréal, Québec: The first paleoseismic trench in eastern Canada. *Seismica*. doi:[10.26443/seismica.v3i2.1179](https://doi.org/10.26443/seismica.v3i2.1179)

2023 **Mark, HF**, Lizarralde, D, Wiens, DA. Constraints on bend-faulting and mantle hydration at the Marianas Trench from seismic anisotropy. *Geophysical Research Letters*. doi:[10.1029/2023GL103331](https://doi.org/10.1029/2023GL103331)
Open data: [scripts for anisotropy calculations](#)

- 2022 Russell, JB, Gaherty, JB, **Mark, HF**, Hirth, G, Hansen, LN, Lizarralde, D, Collins, JA, Evans, RL. Seismological evidence for girdled olivine lattice-preferred orientation in oceanic lithosphere and implications for mantle deformation processes during seafloor spreading. *Geochemistry, Geophysics, Geosystems*. doi:[10.1029/2022GC010542](https://doi.org/10.1029/2022GC010542)
- 2022 Ben Mansour, W, Wiens, DA, **Mark, HF**, Russo, RM, Richter, A, Marderwald, E, Barrientos, S. Mantle flow pattern associated with the Patagonian slab window determined from azimuthal anisotropy. *Geophysical Research Letters*. doi:[10.1029/2022GL099871](https://doi.org/10.1029/2022GL099871)
- 2022 **Mark, HF**, Wiens, DA, Ivins, ER, Richter, A, Ben Mansour, W, Magnani, MB, Marderwald, E, Adaros, R, Barrientos, S. Lithospheric erosion in the Patagonian slab window, and implications for glacial isostasy. *Geophysical Research Letters*. doi:[10.1029/2021GL096863](https://doi.org/10.1029/2021GL096863)
Open data: [velocity and viscosity models](#), and some [cross-correlations](#)
- 2021 **Mark, HF**, Collins, JA, Lizarralde, D, Hirth, G, Gaherty, JB, Evans, RL, Behn, MD. Constraints on the depth, thickness, and strength of the G discontinuity in the central Pacific from S receiver functions. *Journal of Geophysical Research: Solid Earth*. doi:[10.1029/2019JB019256](https://doi.org/10.1029/2019JB019256)
- 2019 Shinevar, WJ, **Mark, HF**, Clerc, F, Codillo, EA, Gong, J, Olive, JA, Brown, SM, Smalls, PT, Liao, Y, Le Roux, V, Behn, MD. Causes of oceanic crustal thickness oscillations along a 74-Myr Mid-Atlantic Ridge flow line. *Geochemistry, Geophysics, Geosystems*. doi:[10.1029/2019GC008711](https://doi.org/10.1029/2019GC008711)
- 2019 **Mark, HF**, Lizarralde, D, Collins, JA, Miller, NC, Hirth, G, Gaherty, JB, Evans, RL. Azimuthal seismic anisotropy of 70 Ma Pacific-plate upper mantle. *Journal of Geophysical Research: Solid Earth*. doi:[10.1029/2018JB016451](https://doi.org/10.1029/2018JB016451)
- 2018 **Mark, HF**, Behn, MD, Olive, JA, Liu, Y. Controls on Mid-ocean Ridge Normal Fault Seismicity Across Spreading Rates From Rate-and-State Friction Models *Journal of Geophysical Research: Solid Earth*. doi:[10.1029/2018JB015545](https://doi.org/10.1029/2018JB015545)

Manuscripts in progress

- Submitted **Mark, HF**, Lizarralde, D, Shillington, D, Cortes Rivas, V, Behn, MD. Along-strike seismic structure of the Andreanof Aleutian Arc Segment, with implications for the formation of continental crust. *Copies available on request*.
- In prep **Mark, HF**, Preine, J, Sager, W, Tominaga, M, Tivey, M. Seismic evidence for Cenozoic volcanic activity in the Western Pacific.
- In prep Preine, J, **Mark, HF**, Tominaga, M, Tivey, M, Sager, W, Hübscher, C, Seidel, E, Hartge, M, Friedrich, A, Schebeck, L. Comparison of high-resolution marine seismic sources from academic offshore surveys.

In review Wu, J-N, Tominaga, M, **Mark, HF**, Zhu, J, Fornari, D, Tivey, M, Lanagan, T, Faluotico, S, Hayden, E, Small, M, Aliod, D, Brady, N, Metzger, A, PFPE. Shipboard Marine Gravimeters Operated on Vessels in the U. S. Academic Research Fleet: Capabilities and Data Quality.

Reviewed editorials

2023 **Mark, HF**, Ragon, T, Funning, G, Hicks, S, Rowe, C, Teplitzky, S, Convers, J, Karasözen, E, Corona-Fernandez, RD. Editorial workflow of a community-led, all-volunteer scientific journal: lessons from the launch of Seismica. *Seismica*. doi:[10.26443/seismica.v2i2.1091](https://doi.org/10.26443/seismica.v2i2.1091)

2022 Rowe, C, Agius, M, Convers, J, Funning, G, Galasso, C, Hicks, S, Huynh, T, Lange, J, Lecocq, T, **Mark, HF**, Okuwaki, R, Ragon, T, Rychert, C, Teplitzky, S, van den Ende, M. The launch of Seismica: a seismic shift in publishing. *Seismica*. doi:[10.26443/seismica.v1i1.255](https://doi.org/10.26443/seismica.v1i1.255)

Funding

2023 – 2024 NSF-OCE: “An assessment of low-temperature, ductile lithospheric deformation using existing broadband seismic data from around South Island, New Zealand”. Award ID: [2316757](https://www.nsf.gov/awardsearch/showAward?AWD_ID=2316757). PI: **Mark, HF**. \$84k to HF Mark

2023 – 2025 NSF-OCE: “Collaborative Research: Resolving the Origin of the Jurassic Quiet Zone”. Award ID: [2221814](https://www.nsf.gov/awardsearch/showAward?AWD_ID=2221814). PI: Tominaga, M, co-PIs: **Mark, HF**; Sager, W. \$827k to WHOI including \$192k to HF Mark

Awards

2023 **Seismica Moment Award**, Seismica journal

2019 **Fossett Postdoctoral Fellowship**, Washington University in St Louis

2019 **AGU Outstanding Student Paper Award (OSPA)**, Tectonophysics

2016 **AGU OSPA**, Tectonophysics

2016 **GeoPrisms OSPA**, honorable mention

2015 **National Science Foundation Graduate Research Fellowship**

2014 **Rosenblith Presidential Fellowship**, Massachusetts Institute of Technology

2014 **Phi Beta Kappa**, Beta of Illinois

2013 **John Crerar Science Writing Prize**, University of Chicago

Teaching Experience

2025 – on **Climate Lab**, American Museum of Natural History, Explainer (volunteer)

2025	Buzzards Bay Term , Gull Island Institute, Natural science faculty
2025	ENVR 1200: Dynamic Earth , Northeastern University, Guest lecturer
2024	Buzzards Bay Term , Gull Island Institute, Natural science faculty
2023	WHOI Software Carpentries Python workshop , Course assistant
2020	EPS 564: Multidisciplinary Study of Subduction Zones , Washington University in St Louis, Guest lecturer
2018	Kaufman Teaching Certificate , MIT Teaching and Learning Lab
2018	WHOI Software Carpentries git workshop , Course assistant
2017	MS-3221: Oceanography , Massachusetts Maritime Academy, Co-Instructor
2016	12.710: Elements of Modern Oceanography , MIT-WHOI Joint Program, TA
2016–2017	WHOI Summer Math Review , Lecturer

Academic Service

2025 – on	LDEO Campus Life Committee , Department representative
2022 – on	Seismica journal management committee, Co-chair of Standards and Copy Editing
2022 – 2024	WHOI Postdoc Association , Department representative
2022 – 2024	WHOI Workplace Climate Committee , Postdoc representative
2022 – 2023	WHOI G&G Proposal Club , Co-organizer
2021	WUSTL EPS Department URGE pod , Pod leader
2018 – 2019	500 Women Scientists of Cape Cod , Steering Committee member
2015 – 2019	MIT-WHOI Joint Program Farrington Collection , Librarian
2014 – 2016	WHOI Summer Math Review , Coordinator
	Peer reviewer for: JGR Solid Earth, Nature Geoscience, Geophysical Research Letters, Science Advances, Nature Communications, Journal of Open Source Software, Geoscience Letters, Earth and Planetary Science Letters, Seismological Research Letters, Geological Society of London Special Publications, Seismica, Tectonics, The Seismic Record, Canadian Journal of Earth Sciences, Geophysical Journal International, Geology
	Conference sessions convened: SSA Annual Meeting 2022, <i>Marine Seismology</i> ; AGU Fall Meeting 2021, <i>EP15 – Coupling of the Cryosphere, Solid Earth, Surface, and Climate in Shaping Late Cenozoic Topography</i>

Workshops and short courses

2022	IRIS Magnetotelluric Instrumentation and Data Processing Short Course
2017	WHOI Communications Course – How NOT to write for peer-reviewed journals: Talking to everyone else
2017	MBL Writing about Science for the Public workshop

2015 WHOI Ocean Law and Policy Seminar

Invited Seminars

2024 **Lamont Doherty Earth Observatory**, MPG/SGT Department seminar
 Texas A&M University, Oceanography Department seminar

2023 **McGill University**, EPS Department seminar
 University of Arizona, Geosciences Colloquium

2022 **University of Washington School of Oceanography**
 University of Wisconsin Madison, Weeks Seminar
 Syracuse University, EES Department Colloquium

2021 **University of Hawai'i Manoa**, Earth Sciences Department seminar

2019 **University of Illinois Urbana-Champaign**, Brown bag seminar

2018 **Lamont-Doherty Earth Observatory**, MGG/SGT seminar
 WHOI, Interdisciplinary Biology Seminar Series on Acoustics

Field Experience

2026 R/V *Thompson*, TN449.
 Multi-scale magnetic survey in the western Pacific with AUV Sentry.

2024 R/V *Sikuliaq*, SKQ202418S.
 MCS and magnetic survey in the western Pacific (chief sci).

2020 R/V *Marcus G. Langseth*, MGL2003.
 Active-source reflection and refraction survey in the Central Aleutians

2017 R/V *Neil Armstrong*, AR23-02.
 Collecting underway data along a ridge flowline in the Atlantic

2016 R/V *Neil Armstrong*, AR05.
 Scientific validation cruise

2015 ENAM land seismic experiment, North Carolina.
 RT-125 deployment and recovery for active-source seismic lines on the Eastern North American Margin

2014 PRIDE SeisORZ seismic experiment, Botswana.
 RT-125 deployment and recovery for an active-source seismic line across the Okavango Delta

Selected Communication and Outreach

2023 – on Essay reviewer for [Syrian Youth Empowerment Initiative](#).

2021 Panel speaker for virtual AGU event on "Submitting your first paper"

2020 – 2022 Contributing writer and peer editor at [geobites.org](#)
 Bloggng, general audience summaries of new research

- 2018 [How is the seafloor made?](#)
Magazine article, *Oceanus* Vol 53. No. 2
- 2017 (Way) under the sea: Imaging the rocks beneath the deep ocean
Public talk, Science Made Public lecture series, Woods Hole, MA
The lithosphere-asthenosphere boundary: What it is, where it is, and why you should care
Public talk, Woods Hole Public Library, Woods Hole, MA
- 2016 Under-under the sea: Imaging the rocks beneath the deep ocean
Public talk, Woods Hole Public Library, Woods Hole, MA

Conference Abstracts (*=invited)

- 2025 **Mark, HF**, Lizarralde, D, Shillington, D, Cortes Rivas, V, Behn, MD. *Seismic structure and estimated composition of arc crust in the Andreanof segment of the Aleutian Arc from wide-angle refraction data*. AGU Fall Meeting, New Orleans, LA.
Mark, HF, Preine, J, Tominaga, M, Sager, W, Tivey, M, Moreno, E. *Multiple generations of magmatic activity imaged by high-resolution seismic reflection data in the western Pacific*. AGU Fall Meeting, New Orleans, LA (poster).
Moss, M, Myers, M, **Mark, HF**, Yang, X, Cheng, Y, Grapenthin, R, Lizik, Y, Portner, DE, Lee, WJD, Wu, S-M, Larsen, JF, Coombs, ML, Freymuller, JT. *Petrologic and Geophysical Synthesis of the Magmatic System at Akutan Volcano*. AGU Fall Meeting, New Orleans, LA.
Mark, HF, Lizarralde, D, Shillington, D, Cortes Rivas, V, Behn, MD. *Seismic Structure of the Andreanof Segment of the Aleutian Arc, and Implications for the Composition of Arc Crust*. Alaska-Aleutian Arc Workshop, Anchorage, AK.
- 2024 **Mark, HF**. *Estimating the Extent of Low-temperature Ductile Deformation in the Lithosphere Using Seismic Anisotropy Measurements Around the Alpine Fault*. SSA Annual Meeting, Anchorage, AK.
Mark, HF, Lizarralde, D, Shillington, D, Cortes Rivas, V. *Seismic Structure of Arc Crust in the Andreanof Segment of the Aleutian Arc from Wide-angle Refraction Data*. SSA Annual Meeting, Anchorage, AK.
***Mark, HF**. *Open Science for geoscience workshop*. ACCESS, Tempe, AZ.
- 2023 Gourdeau, A, Prush, V, Rowe, C, Wang, K, Rosset, P, Chouinard, L, Lamothe, M, **Mark, HF**, Morris, I, Laly, M, Nackers, C. *An Ongoing Search for Active Faults in the Western Quebec Seismic Zone, Eastern Canada*. AGU Fall Meeting, San Francisco, CA.
- 2022 ***Mark, HF**, Collins, JA, Lizarralde, D, Gaherty, JB, Hirth, G, Evans, RL, Behn, MD. *Where is the G discontinuity, what does it represent, and how can we tell? A case study from the NoMelt experiment*. AGU Fall Meeting, Chicago, IL.
Cortes Rivas, V, Shillington, D, Lizarralde, D, **Mark, HF**, Estep, JD, Boston, B. *Seismic reflection imaging of along-strike changes in forearc structure in the Andreanof segment of the Aleutian subduction zone*. AGU Fall Meeting, Chicago, IL.

- Russell, JB, Gaherty, JB, **Mark, HF**, Hirth, G, Hansen, LN, Lizarralde, D, Collins, JA, Evans, RL. *Seismological Evidence for Girdled Olivine Fabric in Oceanic Lithosphere and Implications for Mantle Deformation Processes During Seafloor Spreading*. AGU Fall Meeting, Chicago, IL.
- Wiens, DA, Ben Mansour, W, **Mark, HF**. *Seismic Anisotropy and Mantle Dynamics Associated with Slab Windows*. AGU Fall Meeting, Chicago, IL.
- Mark, HF**, Wiens, DA, Lizarralde, D. *Insights into bend-faulting and mantle hydration at the Marianas trench from seismic anisotropy*. SSA Annual Meeting, Bellevue, WA (poster).
- 2021 **Mark, HF**, Wiens, DA, Ben Mansour, W, Ivins, ER, Richter, A, Magnani, MB, Marderwald, E, Adaros, R, Barrientos, S. *Thermal erosion of the lithosphere in the Patagonian slab window and implications for glacial isostatic adjustment*. AGU Fall Meeting, New Orleans, LA.
- Mark, HF**, Lizarralde, D, Shillington, D, Cortes Rivas, V, Estep, JD. *Crustal structure along-strike in the Andreanof segment of the Aleutian Arc from wide-angle seismic refraction data*. AGU Fall Meeting, virtual (poster).
- Wiens, DA, Ben Mansour, W, **Mark, HF**, Shore, P, Richter, A, Barrientos, S. *Geodynamics of the Patagonian Slab Window Constrained by Shear Wave Splitting and Seismic Imaging*. AGU Fall Meeting, New Orleans, LA.
- Mark, HF**, Wiens, DA, Lizarralde, D. *Estimating bend-faulting and mantle hydration at the Marianas trench from seismic anisotropy*. EGU General Assembly, virtual.
- ***Mark, HF**. *Anisotropy in slightly more than one dimension*. Marine Seismology Symposium, virtual.
- Mark, HF**, Wiens, DA, Lizarralde, D. *Estimating bend-faulting and mantle hydration at the Marianas trench from seismic anisotropy*. Marine Seismology Symposium, virtual (poster).
- 2020 **Mark, HF**, Wiens, DA, Pourpoint, M, Magnani, MB, Ivins, ER, Richter, A, Barrientos, S. *Seismic structure and the extent of the slab window beneath the Northern and Southern Patagonian Icefields*. AGU Fall Meeting, virtual (poster).
- 2019 **Mark, HF**, Collins, JA, Lizarralde, D, Hirth, G, Gaherty, JB, Evans, RL. *The depth, sharpness, and strength of the G discontinuity from S-to-p receiver functions at the NoMelt site in the central Pacific*. AGU Fall Meeting, San Francisco, CA.
- 2018 **Mark, HF**, Collins, JA, Lizarralde, D, Hirth, G, Gaherty, JB, Evans, RL. *S-to-p receiver functions for the central Pacific from NoMelt*. AGU Fall Meeting, Washington, D. C.
- Shinevar, WJ, **Mark, HF**, Clerc, F, Codillo, EA, Olive, JA, Gong, J, Brown, SM, Smalls, PT, Liao, Y, Le Roux, V, Behn, MD. *Temporal variability of seafloor spreading processes documented along an 80-Myr geophysical transect across the Mid-Atlantic Ridge*. AGU Fall Meeting, Washington, D.C.

- Gaherty, JB, Russell, JB, **Mark, HF**, Lin, PP, Sarafian, EK, Ma, Z, Lizarralde, D, Collins, JA, Hirth, G, Evans, RL, Dalton, CA. *A comprehensive portrait of the central Pacific lithosphere-asthenosphere system from NoMelt seafloor geophysical observations*. AGU Fall Meeting, Washington, D.C.
- 2017 ***Mark, HF**, Behn, MD, Liu, Y, Olive, JA. *Geometric and thermal controls on normal fault seismicity*. AGU Fall Meeting, New Orleans, LA.
- Mark, HF**, Lizarralde, D, Collins, JA, Miller, NC, Hirth, G, Gaherty, JB, Evans, RL. *Seismic anisotropy of 70 Ma Pacific-plate upper mantle*. AGU Fall Meeting, New Orleans, LA.
- *Mark, HF**, Behn, MD, Liu, Y, Olive, JA. *Seismic coupling at divergent plate boundaries from rate-and-state friction*. GeoPrisms TEI on Rift Initiation and Evolution, Albuquerque, NM.
- 2016 **Mark, HF**, Behn, MD, Liu, Y, Olive, JA. *Seismic coupling at divergent plate boundaries from rate-and-state friction*. AGU Fall Meeting, San Francisco, CA.
- Mark, HF**, Behn, MD, Liu, Y, Olive, JA. *Rate-and-state friction models of seismic cycles on oceanic normal faults*. GRC on Rock Deformation, Andover, NH (poster).
- Mark, HF**, Lizarralde, D, Gaherty, JB, Collins, JA, Hirth, G, Evans, RL, Lin, PP. *Seismic anisotropy in the Pacific upper mantle*. Seismology Student Workshop, Lamont Doherty Earth Observatory.
- 2014 **Mark, HF**, Lizarralde, D, Gaherty, JB, Collins, JA, Hirth, G, Evans, RL. *Pacific upper mantle seismic anisotropy from the active-source component of the NoMelt experiment*. AGU Fall Meeting, San Francisco, CA (poster).
- 2012 Heinz, DL, **Mark, HF**. *The Effect of Wavelength-Dependent Emissivity on the Melting Temperature of Iron from Shock Wave Measurements*. AGU Fall Meeting, San Francisco, CA (poster).